

Photo Essay: How to Build a Seneca



Pacific Power Dory

Plans for this boat may be found at:
<http://www.spirainternational.com/>



To begin building the Seneca, or any Spira International framed boat, a strongback jig is required. This is simply a sturdy beam on which the frames and other elements of the hull's framing are set up and temporarily attached to hold them in position until the adhesive holding the frame together is cured.

The frames need to be set up on the strongback, in most cases, higher than the strongback to establish the fair curve of the bottom. (Some boats have a completely flat bottom.) The blocks need not be as elaborate as these, but these work very well and make the following tasks much easier.



When the frames are set up on the strongback, it looks something like this.



The keelson is added next to the frames. Note that the cutout in the frames is wider than the keelson. This is intentional, to create limber holes so that bilge water can move back and forth between frames easily.

Notches in the frames are cut and the longitudinal elements are added. After the keelson, the chine log is the first to be installed. It runs the length of the bottom and side joint.



The same procedure used for the chine log is used for the sheer clamp, the longitudinal element running on the top of the frames. (Actually on the bottom in this view because the boat is upside down.)



The boat's framing is "faired," that is, cut to follow the fair curves of the side and bottom panels, so that the plywood planking will have a nice flat surface to lay against and get bonded.

The framing itself is used as a pattern to cut out the plywood. It doesn't really matter if the bottom plywood or side plywood is attached first, but it does matter that whichever is laid first, it must be accurately trimmed to be flush with the framing surface so that the next panel will cover the end grain of the first.



Where the plywood meets, butt blocks made of scrap plywood are bonded to the inside of the joint, and screws run through the outer ply into the butt block to ensure a secure doubled joint.

You can scarf splice the plywood together, but it is a lot more work and not really necessary. These simple butt-joined panels are amply strong for anything you boat is likely to come across.

Here, you can see the meticulous detail this builder used when creating the planking joints on the bow of the boat. Achieving this level of craftsmanship is not difficult, it just requires patience and attention to detail.





Once the planking is on, the hull starts looking like it is really a boat.

Once the epoxy or polyurethane glue sets, the hull becomes rigid and you can loosen or remove it from the strongback jig.

The outside of the hull is prepped by filling any counter-sunk screw holes, any seams at the butt blocks, and any other finish defects. Patience here pays off in the long run.

When this part of the hull is complete, most builders elect to cover the hull with fiberglass and epoxy resin. Two layers of 6 oz glass cloth and epoxy laminating resin are normally recommended.

The hull is then flipped over and removed from the strongback. The inside is saturated with epoxy resin. No fiberglass cloth is needed. The epoxy strengthens and seals the wood making the boat both sturdier and more rot resistant.





Though not specifically called out in the plans, this builder elected to add an inwale on the inside of the frames. This gives the hull a more finished appearance.

Strip decking was added to this hull. Left bright (varnished,) the wood appearance is elegant and traditional. These strips are attached directly to the topside of the lower frame elements.





Next the various accessories are added as desired by the builder. Cleats, motor, oar locks, and in this case swivel seats. finish the hull and make it seaworthy.

Ahhhh... This is what it's all about!

